



SWS3009 · ADVANCE PROPOSAL

# DeskMate

**Team 11 — Loss Hunters**

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A MOVING DESKTOP COMPANION FOR FOCUS, WELL-BEING AND INTERACTION



ADVANCED PROJECT

## DESKMATE DESKTOP COMPANION

10 DAYS

Vision · deep learning · autonomous robotics

- Wang Tianqi · Lin Ziqi
- Gu Songtao · Chen Zengchun
- Loss Hunters · Group 11



### PRODUCT VISION

A desktop companion that sees context  
and responds through motion and expression



### TARGET USERS

Students and office workers  
who stay at a desk for long periods



### CORE PROBLEM

Important habits are easy to miss  
when reminders are passive or easy to dismiss



### DESIGN PRINCIPLE

Proactive, responsive  
and fun enough to stay present on the desk



### COURSE FIT

Deep-learning perception + autonomous robot control  
Video + computer audio · moving parts · deep-learning automation · interactive feedback

FROM PASSIVE REMINDERS TO A COMPANION THAT PERCEIVES AND ACTS



PRODUCT PROMISE

## AWARE OF DESK CONTEXT

### PROACTIVE

Perceive → decide → move and respond

- Reduce unnoticed long sitting
- Interrupt harmful focus patterns
- Make reminders visible and engaging



### LONG SITTING

Detect prolonged seated or static states  
then trigger a stretch or movement cue



### SCREEN & EYE USE

Estimate face-to-screen distance  
and sustained head / gaze direction



### PHONE DISTRACTION

Detect persistent phone use  
during an active focus session



### PASSIVE SOFTWARE

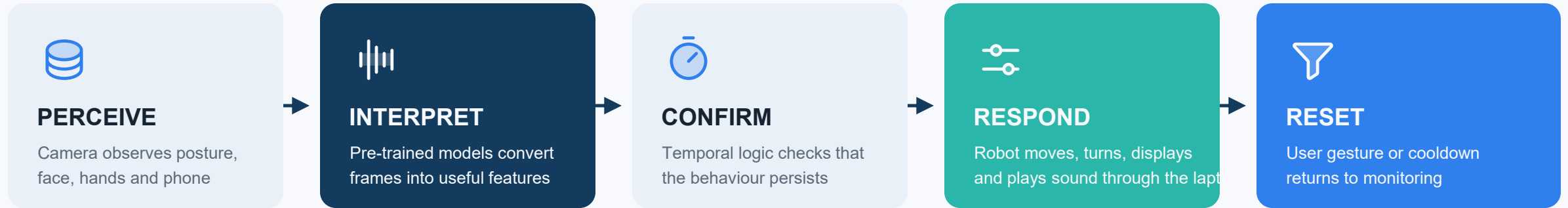
Notifications are easy  
to mute, ignore, or forget



### DESKMATE RESPONSE

One event layer coordinates movement, display, servo and computer sound  
The robot changes position or expression so the reminder becomes a physical interaction

# CORE EXPERIENCE



 **FOCUS SUPPORT**

- Start, pause and reset a focus timer
- Detect persistent phone distraction
- Celebrate completion with motion / display

 **WELL-BEING CUES**

- Track sitting and static duration
- Estimate screen distance and head direction
- Prompt short breaks and posture changes

 **COMPANION FEEDBACK**

- Face recognition can personalize greetings
- Screen shows emoji, status and timer
- Movement makes the response visible and playful

 **ONE LOOP, MULTIPLE MODALITIES**  
Video and laptop audio drive decisions; motion, display and sound close the interaction loop.

# SYSTEM ARCHITECTURE



| PERCEPTION / DECISION                                                                                                                         | ROLE   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|--------|
|  <b>Vision models</b><br>Pose, face, hand and phone          | events |
|  <b>Temporal logic</b><br>Smooth, confirm and apply cooldown | gate   |
|  <b>Runtime choice</b><br>Raspberry Pi 5 or laptop          | deploy |
|  <b>Communication</b><br>USB serial JSON messages          | serial |

| CONTROL / FEEDBACK                                                                                                                          | ROLE     |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------|
|  <b>Drive base</b><br>4 TT motors + HW-130 driver        | motion   |
|  <b>Interaction</b><br>64×48 display + two servos        | feedback |
|  <b>Safety sensors</b><br>Ultrasonic + tracking modules | safety   |
| <b>Scope boundary</b><br>Two-wheel + caster and a mechanical arm remain optional mechanical extensions                                      |          |

## LAYER 01

### PRE-TRAINED PERCEPTION



#### POSE

MediaPipe Pose  
Landmarker



#### FACE / EYE

MediaPipe Face  
Landmarker



#### HAND

MediaPipe / cvzone  
21 landmarks



#### PHONE

Object Detector +  
EfficientDet-Lite0



#### IDENTITY\*

Optional YuNet +  
SFace embedding

## LAYER 02

### DERIVED FEATURES

#### POSTURE

Body landmarks  
Shoulder / neck / trunk angles

#### MOTION

Landmark displacement  
Variance + static duration

#### EYE / HEAD

Eye landmarks + blendshape  
Head pose + distance level

#### GESTURE

Five-finger state  
Direction, speed, trajectory

#### DETECTION

Phone box + confidence  
Identity similarity (optional)

## LAYER 03

### TEMPORAL + RULE LOGIC

#### MOVING AVERAGE

Smooth frame-level features

#### MULTI-FRAME VOTING

Confirm stable state decisions

#### DURATION THRESHOLD

Emit only persistent states

#### REMINDER COOLDOWN

Avoid repeated interruptions

## SUPPORT

### PROGRAM LOGIC



Brightness statistics



Audio level / SPL



Clock and reminders



Robot actuation

# UNIFIED EVENT OUTPUT



## RAW MODEL OUTPUTS

### KEYPOINTS

Body · face · hand

### DETECTION

Class · box · confidence

### IDENTITY

Embedding · similarity

### GESTURE

Class · probability

### AUDIO / STATE

Laptop noise level · state probability



## STANDARDIZED EVENT

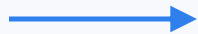
```
{  
  "event": "bad_posture",  
  "model_level": "basic",  
  "confidence": 0.91,  
  "duration_s": 38,  
  "evidence": {  
    "neck_angle_deg": 24.6  
  },  
  "suggested_action": "move_and_prompt"  
}
```

## EVENT VOCABULARY

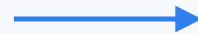
bad\_posture  
static\_too\_long  
screen\_too\_close  
low\_blink\_rate  
environment\_too\_bright  
environment\_too\_dark  
noise\_too\_high\*  
phone\_usage\_detected  
off\_task\_behaviour  
user\_recognized  
unknown\_user  
gesture\_detected



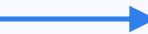
DL MODULE



UNIFIED EVENT



CONTROLLER



# HARDWARE & CONTROL

## COMPUTE

- 1 Raspberry Pi 5 · high-level compute
- 2 Laptop · optional model runtime
- 3 IMX219 camera · video input
- 4 Arduino Mega 2560 motor + servo controller
- 5 USB serial link
- 6 Power bank for Raspberry Pi separate 3S motor pack
- 7 Shared ground + protected wiring

## MOTION

- 1 Four TT geared motors
- 2 HW-130 motor driver
- 3 Four-wheel differential drive primary and safest base
- 4 Two-wheel + caster exploratory configuration



Primary demo uses four-wheel differential drive

## INTERACTION

- 1 64×48 monochrome display
  - 2 MG90S + MG90 servos
  - 3 Ultrasonic + tracking sensors for obstacle and edge awareness
  - 4 Speaker and microphone through laptop if connected
-  Laptop audio supplies the second input modality

### BUILD PRIORITY

### SAFE BASE

4-wheel drive · camera · display

### CORE AI

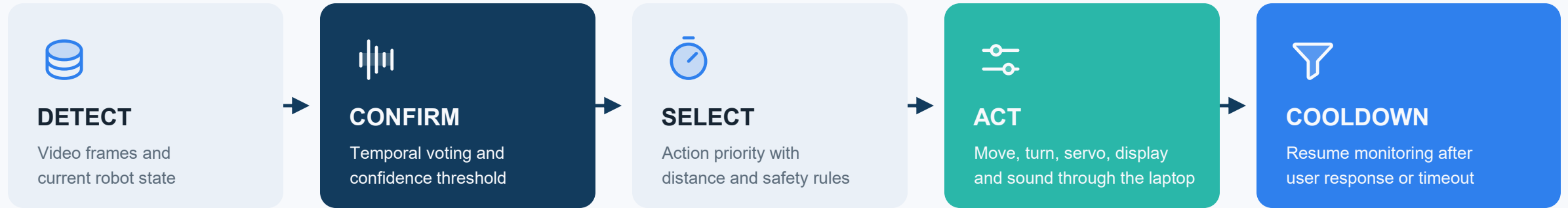
Pose · hand · phone · laptop audio

### EXPLORE

2-wheel base · audio · arm if feasible



# AUTONOMOUS BEHAVIOUR



 **SITTING TOO LONG**

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- Show timer and stretch cue
- Approach or turn toward the user
- Return after acknowledgement

 **PHONE DISTRACTION**

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- Display a refocus prompt
- Move or react when usage persists
- Gesture acknowledges the reminder

 **GESTURE INPUT**

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- Start, pause or reset the focus timer
- Robot confirms with display and motion
- Manual stop remains available

 **AUTONOMOUS DOES NOT MEAN UNCONTROLLED**  
Low speed, distance limits, obstacle sensing and cooldowns remain active during every behaviour.

BUILD CORE FIRST; ADD EXTENSIONS ONLY AFTER THE END-TO-END DEMO WORKS



## DELIVERY STRATEGY

## VERTICAL SLICE THEN POLISH

### MVP FIRST

Every stage ends with a runnable demo

- Must: drive + video + laptop audio
- Should: posture / phone / gesture DL
- Could: identity, 2-wheel base, arm



### DAYS 1-2 · FOUNDATION

Drive, sensors, power and serial link  
Produce a remotely testable moving base



### DAYS 3-4 · PERCEPTION

Camera + pose / face / hand / phone  
Validate frame rate and event quality



### DAYS 5-6 · RESPONSE

Unified events → Arduino actions  
Add timer, display, servo and movement



### DAYS 7-8 · INTEGRATE

Tune thresholds  
and complete end-to-end scenarios



### DAYS 9-10 · TEST + DEMO

Run scenario tests, polish behaviour, prepare slides and backup video  
Failsafe: each AI event can be simulated so hardware and interaction remain demonstrable

# APPENDIX: OPTIONAL EXTENSIONS

ONLY AFTER CORE DEMO

USE ONLY WHEN CONTINUOUS BEHAVIOUR IS NOT STABLE IN THE PRE-TRAINED PIPELINE



RECOMMENDED LEARNED EXTENSION

## DYNAMIC GESTURE TCN OR GRU

$T \times 21 \times 3$

normalized hand-landmark sequence

- Wave, swipe, and circle classes
- Low-dimensional training input
- Lower background and privacy risk



## CONTINUOUS POSTURE

Small MLP, 1D-TCN, or GRU  
Normalized body-landmark sequence



## PHONE DETECTOR

Fine-tune a lightweight detector  
Small targets and partial occlusion



## OFF-TASK BEHAVIOUR

TCN/GRU or multimodal MLP  
Phone, head direction, idle time, events



## AUDIO CLASSIFICATION

Pre-trained YAMNet via laptop  
Requires an external microphone



## PERSONALIZED CALIBRATION

User-specific distance, posture and usage baselines  
Goal: reduce variation between users without making personalization a core dependency